

Exploring drivers, barriers and enablers for manufacturing reshoring in the Nordic countries

Review of
International
Business and
Strategy

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Received 5 November 2024

Revised 4 April 2025

3 May 2025

Accepted 5 May 2025

Abstract

Purpose – This study aims to investigate the dynamics of manufacturing reshoring in Nordic countries by identifying the drivers, barriers and enablers influencing the decisions.

Design/methodology/approach – Grounded in in-depth interviews with 16 Swedish manufacturing firms, this research uses the eclectic paradigm as its theoretical framework.

Findings – This research reveals 65 drivers, 32 barriers and 38 enablers for manufacturers to bring manufacturing back. The primary drivers include total cost, product quality, controllability and flexibility in production. Prominent barriers are labour resources, facilities and equipment, labour, overhead and total costs. Key enablers include manufacturing capacity, labour resources, know-how and intellectual property, infrastructure and automation level.

Research limitations/implications – This research demonstrates how a single decision factor can manifest as a driver, barrier or enabler depending on specific scenarios, adding nuance to the existing reshoring literature. By applying the eclectic paradigm, this research reinforces the applicability of the theory in manufacturing location decision research and integrates factor perspectives with strategic-seeking categories. It refines and encourages the use of the eclectic paradigm.

Practical implications – The research offers practical implications for policymakers, industry leaders and researchers seeking to understand Nordic industrial dynamics. Insights provide guidance for industry leaders in strategic decision-making and navigate policymakers' design measures to support industrial competitiveness in the Nordic region.

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This work was supported by the Knowledge Foundation through the project *Assure: Initial assessment of manufacturing relocation decisions* (Grant 20220013-H-01).

Data availability statement: The participants of this study did not give written consent for their data to be shared publicly, so due to the sensitive nature of the research supporting data is not available.



Review of International Business
and Strategy
Vol. 35 No. 5, 2025
pp. 583-612
Emerald Publishing Limited
2059-6014
DOI 10.1108/RIBS-11-2024-0132

Originality/value – This research is among the few studies integrating strategic advantages with three aspects of the decision factors in analysing manufacturing reshoring. It advances theoretical understanding by showing how the same factor can shift depending on contextual dynamics and serve companies' strategic needs.

Keywords Decision-making, Reshoring, Nordic countries, Drivers, Barriers, Enablers

Paper type Research paper

1. Introduction

Choosing the right manufacturing locations constitutes a vital part of the manufacturing strategy implementation (Mohiuddin *et al.*, 2019; Uluskan *et al.*, 2017). The manufacturing location decision should be aligned with the business strategy and enable the firm to gain competitive advantages in resources, markets, efficiency or strategic assets (Dunning, 1998). In the past decade, the trend in manufacturing relocation has shifted from globalisation (i.e. offshoring) to deglobalisation (i.e. reshoring) (Boffelli *et al.*, 2020; Grappi *et al.*, 2020; D'Ambrosio and Lavoratori, 2025). Similar to offshoring initiatives, companies embracing reshoring strategies have experienced numerous success stories, along with challenges and failures (Engström *et al.*, 2018a, 2018b; Stentoft *et al.*, 2018; Vissak, 2024). Many firms encountered issues with their offshored operations, leading companies to recognise the importance of choosing the “right shoring” location decision (Joubioux and Vanpoucke, 2016). Behind this reshoring phenomenon lies the pursuit of competitive advantages; manufacturing firms have relocated their operations to locations deemed as advantageous for various reasons (Wiesmann *et al.*, 2017; Boffelli *et al.*, 2021; Üner *et al.*, 2023). In light of this trend, the reshoring of manufacturing has attracted considerable attention in recent years from scholars, practitioners and policymakers (Barbieri *et al.*, 2018; Baraldi *et al.*, 2024; Dachs *et al.*, 2019). As compared to research focusing on offshoring, reshoring research is still in its infancy, but growing fast (Li *et al.*, 2025; Pegoraro *et al.*, 2022).

Reshoring activities have predominantly been observed in the USA (e.g. Tate, 2014; Ellram *et al.*, 2013; Chen *et al.*, 2022; Vanchan *et al.*, 2018), as well as in the UK and Western Europe (e.g. Moradlou *et al.*, 2021; Martínez-Mora and Merino, 2020; Srai and Ané, 2016). Research indicates that reshoring helps industrial practitioners strengthen connections with targeted markets and enhances the flexibility of supply chain operations and management (Liu *et al.*, 2023; Lamperti *et al.*, 2025; Wu, 2025). The existing reshoring literature includes three main research streams: decision-making (e.g. Boffelli *et al.*, 2020; Engström *et al.*, 2018a, 2018b; Wiesmann *et al.*, 2017), implementation (e.g. Boffelli *et al.*, 2018; Lund and Steen, 2020) and outcomes (Ulskan *et al.*, 2017; Stentoft *et al.*, 2018). The decision-making stream garnered the most attention to date, with a focus on understanding factors influencing manufacturing reshoring. Specifically, scholars have examined drivers, i.e. factors that trigger reshoring; barriers, i.e. factors that impede reshoring; and enablers, i.e. factors that facilitate reshoring. It is worth noting that extant research tends to investigate these factors separately, thus lacking comprehensive understanding, which emphasises the need for a more systematic and integrative approach.

The Nordic economy is known for its strong emphasis on innovation, sustainability and global competitiveness, with all Nordic countries consistently ranking among the world's most advanced and high-income economies. These countries have shared characteristics, such as high GDP per capita, strong industrial capabilities and a commitment to technological advancement. Among the five countries, Sweden has the largest population 10.54 million (World Bank, 2023), and the GDP per capita was US\$55,516 (World Bank, 2023), around average among them. Sweden also holds a strong global position, ranking

sixth in the IMD World Competitiveness Ranking 2024, second only to Denmark within the region. As such, it serves as a valuable reference point for understanding broader Nordic manufacturing trends. However, with relatively high labour costs, and inflexible labour laws, there was an offshoring trend that encouraged manufacturing firms to establish new capacities in low-cost countries (Mohiuddin *et al.*, 2019). Previous scholars have noticed manufacturing industry within Nordic countries moved parts or all of their production out to low-cost countries from 2010 to 2015 (Johansson and Olhager, 2018a); then, the trend shifted to reshoring. Many companies, such as Ericsson and Sandvik, have now moved some productions back to Sweden because of the digital transformation, high automation level and/or better function synchronisation (Treven, 2022). Especially in recent years, frequent disruptions caused by geopolitical uncertainties and the COVID-19 pandemic have made reshoring a key strategy for manufacturers to enhance resilience and competitiveness (Li *et al.*, 2025; Olhager and Harfeldt-Berg, 2025; D'Ambrosio and Lavoratori, 2025).

So far, limited research has been done in investigating the reshoring drivers, barriers or enablers in the context of the Nordic economy; scholars emphasise the importance of understanding these factors before reshoring becomes a dominant trend in the manufacturing industry (Engström *et al.*, 2018b). Knowledge within this area is rather scattered, thus lacking a systematic view. In addition, a theoretical perspective can enable scholars and managers to develop a profound understanding that is crucial to the decision-making process (Huq *et al.*, 2016). Existing studies have used various theoretical frameworks to understand reshoring, such as transaction cost economics (TCE, Williamson, 1985), the resource-based view (RBV, Barney, 1991) and the eclectic paradigm (Dunning, 1988, 1998). Compared to the other two theories, which focus on cost efficiency and competitive advantage, the eclectic paradigm provides a more holistic framework that integrates ownership, location and internalisation (OLI) advantages. While it considers the interaction of multiple advantages influencing location decisions, the eclectic paradigm remains underutilised in this context. Several scholars argue that this topic requires further investigation (e.g. Engström *et al.*, 2018a, 2018b; Moretto *et al.*, 2020; Pedroletti and Ciabuschi, 2023).

Therefore, this research aims to bridge these research gaps and provide novel insights grounded in theory by leveraging the eclectic paradigm to explore the reshoring drivers, barriers, enablers and interrelations among them, in the Nordic manufacturing context. Two research questions (RQs) were formulated for this purpose:

- RQ1. What are the drivers, barriers and enablers for bringing manufacturing back to Nordic countries?
- RQ2. Based on the eclectic paradigm, what is the linkage among these drivers, barriers and enablers?

The Nordic setting is outlined through a literature review, and the empirical data is provided from a case study conducted in one Nordic country, namely, Sweden. By using the eclectic paradigm as a theoretical lens, this research contributes to both academic and practical decision-making. Through a qualitative approach, data was collected from semi-structured interviews with managers from 16 manufacturing firms with reshoring experience within Nordic countries. The subsequent sections provide a detailed review of existing reshoring literature within the Nordic context (Section 2), research methodology (Section 3), findings (Section 4) and discussions (Section 5). Lastly, Section 6 presents the theoretical contributions, policy implications, limitations and suggestions for future research.

2. Literature review

2.1 Review methodology and samples

A literature review is used to search, select, critically evaluate and synthesise current literature on a particular topic (Cook *et al.*, 1997). Setting clear boundaries is pivotal in conducting a literature review (Ashby *et al.*, 2012). Since around 2009, a growing body of research has concentrated on the phenomenon of reshoring (i.e. the process of bringing manufacturing activities back to the home country) as a response to the limitations of globalisation strategies (Ellram *et al.*, 2013; Tate, 2014). This interest has grown rapidly after the spread of the COVID-19 pandemic, which exposed vulnerabilities in global supply chains and reignited debates around globalisation versus deglobalisation (D'Ambrosio and Lavoratori, 2025; Lamperti *et al.*, 2025). As a result, reshoring has increasingly been examined not only as a strategic location decision or a part of business strategy but also as part of broader economic and political discussions (Pla-Barber *et al.*, 2025; Wu, 2025). This paper set the reshoring background within the Nordic context, which is known for its strong emphasis on innovation, sustainability and global competitiveness. A literature review will benefit in investigating the reshoring of decision-making content within Nordic countries and enable the authors to discover new findings. Thus, the selection criteria for literature include studies being contextualised within Nordic countries; primarily or partially exploring the reshoring decision-making content, such as criteria, factors, drivers, barriers or enablers; and exclusion of literature review studies to ensure original empirical contributions.

In the initial screening step, titles, abstracts and keywords were read to identify papers discussing reshoring to the Nordic countries, 28 relevant papers were identified. After a full paper review, eight papers were excluded, and one paper was added from the search alert. The final sample comprised 21 papers (Table 1), which were included in the full review to reveal drivers, barriers and enablers. Most of the papers within the review sample discussed solely reshoring activities, while nine papers extended the discussion to the manufacturing relocation level, which includes both offshoring and reshoring to extend the knowledge (e.g. Gylling *et al.*, 2015; Heikkilä *et al.*, 2018a, 2018b; Johansson and Olhager, 2018a, 2018b; Olhager and Harfeldt-Berg, 2025). Geographically, 12 studies concentrated on Swedish manufacturing. Two research papers are individually dedicated to Danish, Norwegian and Finnish manufacturing contexts. Furthermore, three papers framed their research within the broader Nordic countries' context.

Reshoring is a relatively new research trend, with nine papers focused primarily on introducing the trend, context and decision-making tools (e.g. Heikkilä *et al.*, 2018a, 2018b; Johansson and Olhager, 2018a; Leisner and Nielsen, 2019), while rest explored factors that affect reshoring decisions. It is worth noting that scholars investigate factors to various extents. For example, Engström *et al.* (2018a, 2018b) and Sequeira *et al.* (2021, 2023) screen the decision-making content to provide a comprehensive view, whereas Fjellström *et al.* (2023), Gylling *et al.* (2015) and El-Sahli and Gullstrand (2023) concentrated on just one or two factors. Interestingly, while research widely discussed reshoring drivers, limited studies explore other roles such as barriers and enablers. Several studies suggested that quality issues experienced in the offshoring location, such as product, infrastructure and service quality, are continuously mentioned as strong reshoring drivers (Engström *et al.*, 2018a, 2018b; Lund, and Steen, 2020). Other strong drivers are delivery lead-time, manufacturing costs and production flexibility (Hilletoft *et al.*, 2021; Olhager and Harfeldt-Berg, 2025; Sequeira *et al.*, 2023). Barriers such as ownership-related issues, high labour costs and raw material costs were mentioned (Arlbjørn and Mikkelsen, 2014; Engström *et al.*, 2018a; Heikkilä *et al.*, 2018a). Additionally, enablers, such as reliable supply networks, efficient external communications with business partners and knowledge transfer, have also been identified

Table 1. Summary of manufacturing relocation studies that explore decision-making content in Nordic countries context

No.	Source	Country	Relocation direction	Research method	Decision factors	Highlights
[1]	Arlbjørn and Mikkelsen (2014)	Denmark	Reshoring	Survey	Drivers and barriers	This note highlights the importance of the use of automation and ambidexterity of globalisation strategies based on data from questionnaires of 843 Danish manufacturing firms
[2]	El-Sahli and Gullstrand (2023)	Sweden	Reshoring	Archival	Drivers	This paper examined two drivers: automation and labour cost. The findings prove technical changes are crucial but not labour costs in reshoring decisions, data were from 1997–2014
[3]	Engström <i>et al.</i> (2018a)	Sweden	Reshoring	Case study	Drivers and barriers	The study analyses data from nine interviews with four case companies, identifies research gaps and explores Swedish reshoring drivers and barriers
[4]	Engström <i>et al.</i> (2018b)	Sweden	Reshoring	Case study	Drivers and barriers	The study analyses data from nine interviews with four case companies and explores Swedish reshoring drivers and barriers based on five categories
[5]	Eriksson <i>et al.</i> (2021)	Sweden	Reshoring	Case study	Enablers	This study uses data from four case companies to explore prerequisites and contextual factors that enable Swedish companies to reshore
[6]	Fjellström <i>et al.</i> (2023)	Sweden	Offshoring and reshoring	Case study	Barriers and enablers	This study explores how knowledge management acts as both a barrier and an enabler in manufacturing relocation strategies, using nine interview data from one Swedish company
[7]	Gylling <i>et al.</i> (2015)	Finland	Offshoring and reshoring	Action research	Driver	This paper examines cost-related factors as relocation drivers for the Nordic bicycle industry
[8]	Heikkilä <i>et al.</i> (2018a)	Finland	Offshoring and reshoring	Survey	Drivers and enablers	This paper reveals the extent, drivers and pattern of manufacturing relocation in Finland, with survey data from 229 Finnish manufacturing firms

(continued)

Table 1. Continued

No.	Source	Country	Relocation direction	Research method	Decision factors	Highlights
[9]	Heikkilä <i>et al.</i> (2018b)	Denmark, Finland and Sweden	Offshoring and reshoring	Survey	Drivers	This paper investigates Nordic manufacturing firms' relocation practices and reveals the trend, extent, drivers and locations with survey data of 847 Nordic manufacturing firms
[10]	Hillierofth <i>et al.</i> (2019)	Sweden	Reshoring	Modelling	Decision criteria	This paper investigates and implements the use of fuzzy rules in making reshoring decisions while testing with some reshoring drivers and testing with ten scenarios
[11]	Hillierofth <i>et al.</i> (2021)	Sweden	Reshoring	Modelling	Decision criteria	This paper tests the suitability of applying the two fuzzy-logic-based support tools in the initial screening of manufacturing reshoring decisions while testing some reshoring drivers
[12]	Johansson and Olhager (2018A)	Sweden	Offshoring and reshoring	Survey	Drivers	Present the drivers of Swedish offshoring and reshoring, with data from 373 surveys conducted between 2010 and 2015
[13]	Johansson and Olhager (2018B)	Sweden	Offshoring and reshoring	Survey	Drivers	Present and compare the location factors and performance of Swedish offshoring and reshoring activities, with 133 offshoring and 99 reshoring survey data
[14]	Johansson <i>et al.</i> (2019)	Denmark, Finland and Sweden	Reshoring	Survey	Drivers	Present and compare the drivers, relationships and benefits of offshoring and reshoring with perspective from factor bundles, data from 275 offshoring and 160 reshoring data
[15]	Leisner and Nielsen (2019)	Denmark, Finland and Sweden	Offshoring and reshoring	Secondary	Drivers	This paper analysed secondary interview data from Heikkilä <i>et al.</i> (2018b) to provide the trend, context and drivers for off- and reshoring cases of the Nordic surface finishing industry

(continued)

Table 1. Continued

No.	Source	Country	Relocation direction	Research method	Decision factors	Highlights
[16]	Lund and Steen (2020)	Norway	Reshoring	Case study	Drivers	This paper presents the drivers for reshoring in Norway and the applicability of the GPN framework for analysing the phenomenon, using 11 semi-structured interview data
[17]	Nuijen et al. (2019)	Norway	Reshoring	Case study	Enablers	This study sheds light on reshoring enablers and discusses how these enablers affect transitions, with 16 interview data from two case companies
[18]	Olhager and Harfeldt-Berg (2025)	Sweden	Offshoring and reshoring	Survey	Drivers	This study is a longitudinal survey study that captures offshoring as well as backshoring before and during the pandemic, investigates how the COVID-19 pandemic changed relocation behaviour
[19]	Sequeira et al. (2021)	Sweden	Reshoring	Modelling	Decision criteria	This study examines six high-level decision criteria with analytical hierarchy process (AHP)-based tools for the initial screening of manufacturing reshoring decisions, testing in 20 scenarios
[20]	Sequeira et al. (2023)	Sweden	Offshoring and reshoring	Modelling	Decision criteria	This study primarily investigates the application of the hybrid fuzzy-AHP-TOPSIS model in evaluating manufacturing relocation decisions, while testing with some reshoring criteria
[21]	Stentoft et al. (2016)	Denmark	Offshoring and reshoring	Survey and case study	Drivers and barriers	This paper presents relocation drivers and barriers in the Danish context, with 245 survey data and interviews of two case companies, with a focus on flexibility as the key factor
Source(s): Authors' own work						

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(Baraldi *et al.*, 2024; Fjellström *et al.*, 2023; Nujen *et al.*, 2019). Due to limited research, conclusions regarding the strong barriers and enablers for firms to come back to Nordic countries have not yet been established. The analysis of drivers, barriers and enablers is presented in Section 2.3, after the introduction of the theoretical lens in Section 2.2.

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2.2 Theoretical lens: the eclectic paradigm
Reshoring decision factors are not decision factors in isolation. The decision factors are related to something that the company wish to, for example, achieve, avoid or benefit from. Examples of similar approaches include research that has focused on reshoring using operations capabilities (Theyel and Hofmann, 2021). In this paper, we join a stream of previous research using the eclectic paradigm trying to understand reshoring. Our approach is centred on exploring a two-dimensional view of decision factors, where they are both categorised into different types of decision factors and how the decision factors can be understood using the eclectic paradigm.

Evaluation through a theoretical lens is pivotal for constructing robust knowledge (Boffelli *et al.*, 2020; McIvor and Bals, 2021; Moradlou *et al.*, 2021). For this reason, the eclectic paradigm (Dunning, 1998) was used to structure the findings. The initial version was OLI parameters, which proposed to uncover the underlying motives that influence manufacturers’ production decisions. The ownership advantage was further divided into the asset (Oa) and transaction (Ot) advantages of international companies (Dunning, 1983). Three main forms of international manufacturing: market-seeking, resource-seeking and efficiency-seeking were introduced (Dunning, 1988). With the dynamic global market situation, the next version of the eclectic paradigm (Dunning, 1998) highlights the growing importance of intangible assets (i.e. knowledge-intensive assets) in the wealth-creating process, as well as emphasises the changing role of company-specific location-bound assets. The “strategic asset-seeking” was added as one important motive of international manufacturing. Although this theory was developed to explain the expansion of international companies, as Fraticchi *et al.* (2016) pointed out, it has also been applied to the manufacturer’s global reconfiguration (e.g. offshoring and reshoring). Therefore, the eclectic paradigm has been used for analysing the determinants of international manufacturing and categorises factors into resource-seeking, market-seeking, efficiency-seeking and strategic asset-seeking:

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- Resource-seeking category: includes factors essential for accessing and leveraging critical resources, including those exclusive to particular locations, more accessible or of superior quality.
 - Market-seeking category: includes factors that exploit market opportunities within specific geographic locations.
 - Efficiency-seeking category: includes factors that explore cost-efficient or productivity-enhancing manufacturing.
 - Strategic asset-seeking category: includes factors that expedite a company’s access to, creation of or maintenance of strategic or knowledge-related assets.

Additionally, factors that fall into multiple categories are classified as hybrid. In this research, the eclectic paradigm is used as the theoretical lens to understand manufacturing reshoring decision-making content. The decision-making content is constructed from decision factors, and some of the decision factors can be further identified as drivers, barriers or enablers (Figure 1). Surprisingly, only a handful of reshoring studies have used this

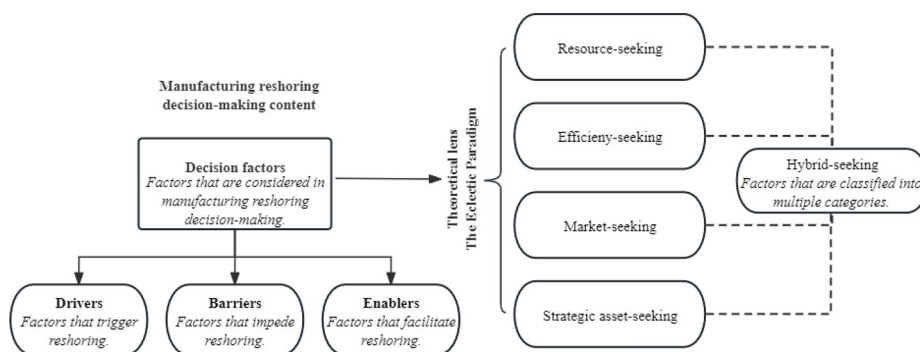


Figure 1. The conceptual framework in this research
Source: Authors' own work

paradigm to understand reshoring decision-making (Ancarani *et al.*, 2015; Barbieri *et al.*, 2018; Ellram *et al.*, 2013; Moradlou *et al.*, 2021).

2.3 Reshoring drivers, barriers and enablers in the Nordic context

To make well-informed decisions, companies need to develop a comprehensive understanding of what and how factors trigger, hinder or facilitate reshoring decisions (Engström *et al.*, 2018a; Presley *et al.*, 2016; Stentoft *et al.*, 2018); these factors are transferable based on specific cases (Engström *et al.*, 2018a, 2018b). Existing reshoring research in the Nordic context (e.g. Heikkilä *et al.*, 2018a; Lund and Steen, 2020; Johansson *et al.*, 2019) has explored drivers, identifying 41 drivers that trigger companies to bring manufacturing back to Nordic countries. By contrast, only a limited amount of research has investigated reshoring barriers (e.g. Engström *et al.*, 2018a, 2018b) and enablers (e.g. Eriksson *et al.*, 2021; Fjellström *et al.*, 2023; Nujen *et al.*, 2019), while understanding barriers and enablers are crucial for effective reshoring implementation. In total, 14 barriers and nine enablers were identified (Table 2).

Within the driver category, 41% of drivers ($n = 17$) aim to enhance cost-efficiency or productivity, followed by 22% ($n = 9$) of drivers seek to enhance or protect companies' strategic assets, 19% ($n = 8$) market-seeking drivers, 12% ($n = 5$) resource-seeking drivers and 5% ($n = 2$) hybrid-seeking drivers.

The decision to reshore manufacturing in the Nordic context is shaped significantly by challenges encountered during offshoring are referred to as “push” drivers (Hilletoft *et al.*, 2021). Notably, within the efficiency-seeking category, “delivery lead time” emerged as a prominent driver. Companies face long lead time due to long distances between factories and markets, which also leads to communication challenges, exacerbated by time differences and language barriers (Engström *et al.*, 2018b; Sequeira *et al.*, 2021, 2023). Additionally, production in foreign countries increased exposure to manufacturing and supply chain risks (Heikkilä *et al.*, 2018a, 2018b; Johansson and Olhager, 2018a, 2018b). The diminishing gap in manufacturing costs between the two countries further motivated companies to reshore (Engström *et al.*, 2018a). Another critical push driver was “product quality”, the negative impact of poor and unstable product quality from offshoring, damaging brand image and reputation. This issue led to the waste of time and resources (Leisner and Nielsen, 2019; Johansson and Olhager, 2018a; Lund and Steen, 2020).

Table 2. Reshoring drivers (D), barriers (B), enablers (E) categorised under the eclectic paradigm

EP	Decision factors	D	B	E	References
Resource-seeking	Business partners	X		X	[7], [16], [17]
	Currency exchange rate	X			[8], [9], [7], [12], [13], [14], [18], [21]
	Facilities and equipment	X			[8], [9], [12], [16], [21]
	Labour resources	X	X		[1], [3], [8], [9], [12], [14], [16], [18]
	Production foundation		X		[21]
	Raw material access and cost	X	X		[3], [4], [8], [9], [12], [13], [16], [18], [21]
	Supply networks			X	[17]
	Customer demand	X			[7], [8], [9], [12], [14], [18], [21]
	Customer service	X			[4]
	Labour cost	X			[8], [9], [12], [13], [14], [18], [21]
Market-seeking	Logistics cost	X			[8], [9], [12], [13], [14], [16], [18], [21]
	Logistics performance	X			[3], [4], [15]
	Macroeconomics	X			[3], [4], [16]
	Market opportunities	X			[3], [4], [16]
	Responsiveness to market	X			[8]
	Coordination cost	X			[16]
	Customer proximity	X			[8], [9], [12], [13], [14], [18], [21]
	Delivery lead time	X			[1], [6], [8], [9], [10], [11], [12], [13], [14], [16], [18], [19], [20], [21]
	External communication	X	X	X	[1], [3], [4], [15], [16]
	Function synchronisation	X			[3]
Efficiency-seeking	Geographical and cultural distance	X			[4], [16]
	Government incentives	X			[3], [4]
	Information technology (IT)		X		[4]
	Labour market flexibility	X	X		[3], [19], [20], [21]
	Legislation and regulations	X	X		[3], [4], [8], [9], [12], [13], [14], [21]
	Manufacturing capacity	X	X	X	[3], [4], [5], [16], [17]
	Manufacturing cost	X		X	[5], [8], [9], [10], [13], [16], [14], [19], [20], [7], [21]
	Manufacturing risks	X			[3], [4], [8], [9], [12], [13], [18], [21]
	Political stability	X			[4]

(continued)

Table 2. Continued

EP	Decision factors	D	B	E	References
Strategic asset-seeking	Production flexibility	X			[8], [9], [10], [11], [12], [14], [16], [18], [19], [20], [21]
	Subsidies	X			[9], [12], [21]
	Supply chain risks	X			[4], [8], [9], [7], [12], [19], [20]
	Time to market	X			[8], [9], [12], [13], [18], [21]
	Brand image and reputation	X			[19], [20]
	Core competencies	X	X	X	[1], [3], [8], [9], [12], [13], [14], [16], [17], [18], [21]
	Innovation ability	X			[11], [19], [20]
	Know-how and IP	X		X	[8], [9], [12], [13], [14], [15], [17], [18]
	Knowledge and technology	X		X	[2], [6], [8], [9], [12], [13], [14], [16], [17], [21]
	Knowledge transfer		X	X	[6], [17]
Hybrid-seeking	Made-in effect	X	X		[3], [4]
	Ownership-related issues		X		[3], [4]
	Product quality	X	X		[1], [3], [4], [7], [8], [9], [10], [11], [12], [15], [16], [18], [21]
	Production and process quality	X			[19], [20]
	Responsible supply chain		X		[4]
	Sustainable supply chain	X			[3], [10], [11]
	Manufacturing automation level	X			[1], [3], [4]
	Research and development (R&D)	X			[9], [12], [13], [14], [16], [18], [21]
	Source(s): Authors' own work				

Specific resources and advantages in home countries that encourage companies to reshore are referred to as “pull” drivers (Hilletoft *et al.*, 2021). For instance, advanced knowledge and technology across various industries in the Nordic countries (El-Sahli and Gullstrand, 2023; Lund and Steen, 2020; Johansson *et al.*, 2019), coupled with strong innovation capabilities (Sequeira *et al.*, 2021, 2023), encourage reshoring. Available manufacturing facilities and equipment proved vital production resources, saving significant investments (Lund and Steen, 2020; Stentoft *et al.*, 2016). Moreover, favourable or stable currency transactions (Johansson *et al.*, 2019; Gylling *et al.*, 2015) and proximity to target markets (Lund and Steen, 2020; Johansson *et al.*, 2019) motivated reshoring. Two drivers served as both strategic asset- and efficiency-seeking advantages: high automation in home countries (Arlbjørn and Mikkelsen, 2014; Engström *et al.*, 2018a, 2018b) and proximity to R&D (Johansson *et al.*, 2019; Stentoft *et al.*, 2016), allowing faster production development (Lund and Steen, 2020).

Barriers, on the other hand, often seen as challenges or issues, hinder reshoring efforts. In the current literature discussing reshoring barriers, strategic asset-seeking accounted for the majority (43%, $n = 6$) of barriers, followed by efficiency-seeking (36%, $n = 5$). Three barriers fell under resource-seeking (21%), with no barriers identified in market-seeking or hybrid-seeking categories.

From a strategic asset-seeking perspective, factors hindering the reshoring process include challenges related to preserving companies’ or manufacturing facilities’ ownership (Engström *et al.*, 2018b) and the potential loss of competencies developed during the offshoring stage (Engström *et al.*, 2018a). Additionally, issues associated with maintaining consistent knowledge management and transfer across borders pose significant barriers (Engström *et al.*, 2018a; Fjellström *et al.*, 2023). The “Made-in effect” proved to have negative impacts, particularly when customers expected lower prices, putting the company at a disadvantage (Engström *et al.*, 2018b). Furthermore, companies often shaping their responsibility image hinder the reshoring process. Some owners delay reshoring initiatives to prevent unemployment among employees at offshored locations due to a sense of social responsibility (Engström *et al.*, 2018b).

From the efficiency-seeking perspective, regulations concerning investments and trade barriers presented challenges (Heikkilä *et al.*, 2018a; Engström *et al.*, 2018b). Communication difficulties with external partners arose during implementation (Engström *et al.*, 2018b). Negative reactions among domestic employees in the host country, a consequence of cutting offshore production, indicated that “labour market flexibility” hinders reshoring (Engström *et al.*, 2018a). Additionally, IT integration proved to be a challenge, as finding suitable solutions to integrate systems implemented in previous production lines was difficult (Engström *et al.*, 2018b). From a resource-seeking perspective, challenges in finding suitable resources in Nordic countries emerged. High labour costs and the scarcity of labour resources with specific competencies were particularly noted (Engström *et al.*, 2018a).

Enablers, often referred to as prerequisites, are essential factors that ensure the success of reshoring initiatives and facilitate their implementation (Eriksson *et al.*, 2021; Fjellström *et al.*, 2023). We identified four strategic asset-seeking enablers (50% of the total), two resource-seeking and two efficiency-seeking enablers (25% of the total).

Efficiency-seeking enablers focus on ensuring the smooth and efficient execution of reshoring efforts. Effective communication with diverse business partners at each step of the reshoring process is vital to prevent unnecessary issues and challenges (Engström *et al.*, 2018b). Moreover, the establishment of parallel manufacturing capabilities and an increase in production capacity significantly enhance the success of reshoring initiatives (Eriksson

et al., 2021; Nujen *et al.*, 2019). As many companies conduct cost-estimation modelling before and during the reshoring process, reasonable manufacturing costs enable companies to continuously process reshoring (Eriksson *et al.*, 2021). From a strategic asset-seeking perspective, intangible resources such as specialised know-how, specific production skills and comprehensive knowledge play a pivotal role. These resources are essential for companies to maintain the same level of product quality and production standards after reshoring (Nujen *et al.*, 2019; Fjellström *et al.*, 2023). Fjellström *et al.* (2023) emphasised the significance of knowledge transfer in creating competitive advantages during reshoring. Insufficient knowledge transfer may lead to reshoring initiatives falling short of their objectives and encountering challenges.

3. Methodology

This research delves into the drivers, barriers and enablers of reshoring in the Nordic manufacturing context. Based on the theoretical framework, we use an inductive research approach inspired by Eriksson and Engström (2021). The research is exploratory in its nature, which is suitable when probing into underdeveloped phenomena (Malhotra and Grover, 1998). Despite some researchers exploring the Nordic reshoring phenomena, there remains a significant gap in understanding the drivers, barriers and enablers simultaneously.

To address this gap, the research methodology used semi-structured interviews for data collection, reporting data captured from case studies. This choice was made due to the method's capacity for facilitating an in-depth exploration of the subject matter. The case selection involved four steps and encompassed multiple studies, allowing for a larger number of cases than what is typically considered manageable in case-study research (Yin, 2014; Eisenhardt, 1989). Understanding Nordic manufacturing reshoring can start with Sweden's representation. Sweden's economic and technological advancements are highly developed and hold a significant position within the Nordic region (Hilmola *et al.*, 2025; Johansson and Olhager, 2018a). Similarities among Nordic economies also provide a basis for analysing the Nordic reshoring phenomenon with Swedish manufacturing as a representative case. Thus, analysing Swedish manufacturing cases provides valuable insights into the broader Nordic manufacturing landscape, serving as an essential starting point for studying manufacturing reshoring within the region. This approach, being a breadth study as supported by Voss *et al.* (2002), allows a deeper exploration of the complexities in manufacturing reshoring decision-making within the Nordic context. However, this approach limited the research exploring the phenomenon in depth, which is a common limitation of breadth studies (Thomas, 2021). Future studies can expand on this research through single-case study to provide more detailed insights and enhance the understanding of reshoring within the Nordic manufacturing context.

3.1 Sample selection

The selection of participants was guided by Four criteria:

- (1) companies that reshored manufacturing were selected;
- (2) respondents from within the companies were required to have experiences in manufacturing reshoring projects;
- (3) the respondent should have experiences in the reshoring decision-making; and
- (4) the respondents' respective companies had to either be located in Nordic countries or have a significant focus on the Nordic manufacturing industry.

Through these criteria, the study endeavoured to gather insights from individuals possessing firsthand knowledge and expertise in the area under investigation. This approach was important, as firsthand experience was anticipated to yield invaluable perspectives on the complexity of manufacturing reshoring decision-making.

Respondents were selected through a two-step process, guided by a dual focus. Initially, companies engaged in reshoring initiatives were recruited. Subsequently, within these companies, suitable respondents were identified to ensure diversity within the company. This step involved discussions with senior management at the respective company to ascertain appropriate candidates. This purposive sampling approach, as outlined by Yin (2014), aimed to gain insights from companies with specific experience. A total of 16 experts were selected.

The selected interviewees comprised seven managing directors and nine production managers, thereby ensuring a balanced representation of top-level production decision-makers. It is noteworthy that all the participating firms were headquartered in Sweden and oriented their operations towards the European market. For a comprehensive overview of the sample, fundamental information about each firm is meticulously detailed in Table 3.

3.2 Data collection

The interviews were conducted using a semi-structured approach, which included open-ended questions related to manufacturing reshoring decision drivers, barriers and enablers. Then, both “what” and “how” questions concerned with the reshoring decision-making drivers, barriers and enablers were asked. The interview protocol was designed in English and translated into Swedish. Participants were given the choice of language, and all interviewees opted for Swedish, though some discussions included English terms. Each interview lasted 50–120 min, allowing enough time to collect sufficient data while also allowing for the flexibility to investigate questions at a deeper level. These interviews were conducted in person in 2018 at the participants’ workplace, ensuring a conducive environment for discussions. All conversations were recorded and fully transcribed. The responses were subsequently translated into English by a bilingual author to ensure accuracy.

Table 3. Interview summary and participant characteristics

ID	Position	Organisation	Location	Interview type	Interview duration
01	Managing director	MC01	Sweden	In-person	106 min
02	Managing director	MC02	Sweden	In-person	102 min
03	Managing director	MC03	Sweden	In-person	110 min
04	Managing director	MC04	Sweden	In-person	107 min
05	Production manager	MC05	Sweden	In-person	70 min
06	Production manager	MC06	Sweden	In-person	90 min
07	Managing director	MC07	Sweden	In-person	70 min
08	Managing director	MC08	Sweden	In-person	102 min
09	Production manager	MC09	Sweden	In-person	120 min
10	Production manager	MC10	Sweden	In-person	88 min
11	Production manager	MC11	Sweden	In-person	88 min
12	Production manager	MC12	Sweden	In-person	52 min
13	Production manager	MC13	Sweden	In-person	72 min
14	Production manager	MC14	Sweden	In-person	50 min
15	Production manager	MC15	Sweden	In-person	73 min
16	Managing director	MC16	Sweden	In-person	93 min

Source(s): Authors’ own work

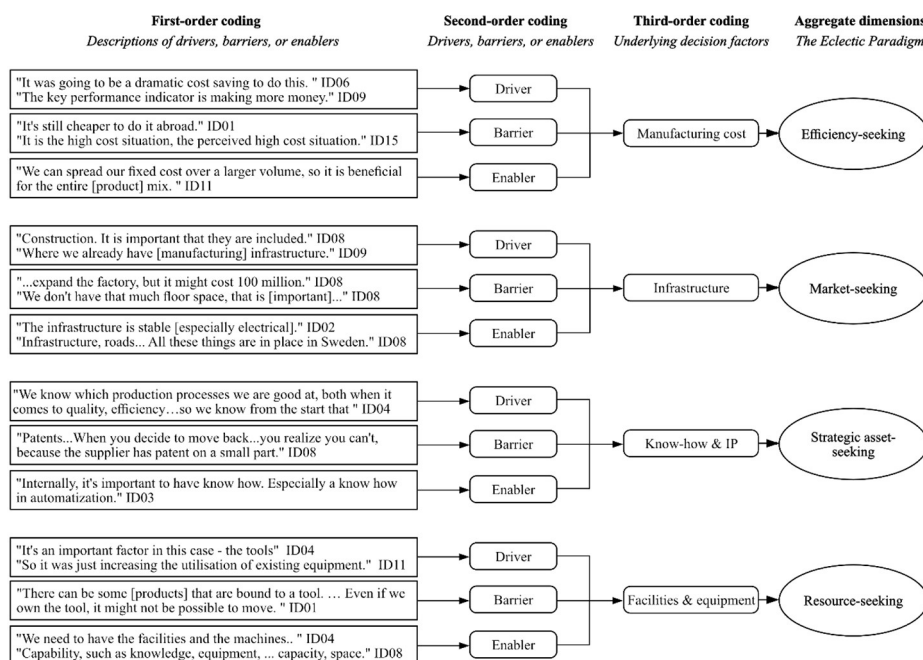


Figure 2. Four-level data coding scheme
Source: Authors' own work

3.3 Data analysis

The interview data analysis used a comprehensive four-level coding scheme (Figure 2). Quotes were analysed by multiple researchers working on larger projects about reshoring. Initially, researchers classified each quote. Quotes related to factors that trigger, impede and facilitate manufacturing reshoring decisions were extracted and constituted the first-order coding, as depicted in Figure 2 under the section titled "Description of drivers, barriers or enablers". In the second step, one researcher took responsibility for each category of quotes. Inductive categories were created according to the principles of open coding (Ellram, 1996). This step was done in an iterative manner, where previously created categories could be removed, renamed or merged. This coding also ensured that the initial classification of quotes was correct, as it is very difficult to put a, for example, driver quote in a specific group of drivers unless the quote really is a driver. Quotes were then coded into the second-level category of "drivers, barriers or enablers". Subsequently, in the third level of coding, these drivers, barriers and enablers were further intricately categorised into underlying decision factors, aiming for a nuanced understanding of the intricacies involved. Finally, an additional level of analysis was introduced using the eclectic paradigm, wherein these factors were classified into efficiency-, market-, strategic asset- or resource-seeking dimensions. The coding process (inspired by Magnani and Gioia, 2023) was then done where all the groups were coded based on theory, allowing for a comprehensive and systematic exploration of reshoring decisions from multifaceted aspects. All the coding was done using spreadsheets.

The data analysis process included two phases. In the first phase, the frequency of each factor mentioned by the interviewees was counted to show, how common the factor was

triggering, impeding or facilitating the Nordic reshoring activities. In the second phase, in each category, the underlying decision factors and related sought advantages were further analysed. This thorough analysis was conducted by the authors with careful consideration to ensure validity and reliability.

4. Findings

A total of 489 usable quotes were coded and analysed. The majority of unrevealing drivers of reshoring (70% of the total, $n = 342$). The remaining were evenly split, unrevealing enablers ($n = 75$) and barriers ($n = 72$) of reshoring. In total, 65 drivers, 32 barriers and 38 enablers of reshoring decisions were identified in the context of Nordic manufacturing (Table 4). The eclectic paradigm is used to categorise drivers, barriers and enablers.

4.1 Reshoring drivers

When discussing the drivers behind reshoring decisions, managers frequently highlighted the efficiency-seeking motives. These included reducing total cost (mentioned in 45 quotes), shortening delivery lead time (mentioned in 33 quotes), enhancing the controllability of production (mentioned in 32 quotes) and improving production flexibility (mentioned in 29 quotes).

Managers often cited drivers that trigger reshoring decisions with a mix of multiple seeking advantages. About 24% of the drivers ($n = 16$) were identified as hybrid-seeking. For example, some companies reshored to access specific know-how or intellectual properties (IPs) in Nordic countries (ID4 and ID16), which reflects a resource-seeking behaviour. Meanwhile, others reshore aimed to protect maintain or safeguard sensitive IPs (ID1, ID8 and ID15), demonstrating an efficiency-seeking strategy. Additionally, some companies reshored to use or further develop their own IPs and production-related know-how (ID3, ID4, ID5, ID6, ID13 and ID16), indicating a strategic asset-seeking approach. Another hybrid-seeking driver is “made-in-effect”, which boosts companies’ brand image. Managers stated that moving back to Sweden strategically improved their brand image through the “Made-in-Sweden” label (ID1, ID4 and ID15). In some cases (ID1, ID3, ID8, ID12 and ID16), managers used this to attract specific customer groups. One manager (ID16) remarked: “For some products you can increase the price if you produce it in Sweden”; another (ID1) added: “Swedish customers want to buy from Swedish companies that buy their inputs in Sweden if the price is not too high”.

Companies displayed a similar interest in exploiting market opportunities and enhancing particular strategic assets, each comprising 14% of the total drivers ($n = 9$). Managers (ID1, ID7 and ID12) stated bringing manufacturing back allowed them to enhance “logistics performance”. One manager (ID10) explained why shortening logistics handling time, and increasing logistics precision are important for accessing market opportunities: “We had large problems [...] long delivery times—6 weeks from China [...] we could not have a high delivery precision. We could not satisfy customer requests”. Additionally, managers stated providing in-time and on-site services to customers increase the competitiveness (ID3, ID4, ID8, ID10 and ID12). Managers also highlighted they reshore to create more value-adding activities for customers (ID4, ID5, ID7, ID8 and ID10), to increase the attractiveness to customers. This is particularly due to the significant potential of the after-sale market in Nordic countries. One manager (ID7) stated: “... something that makes us more attractive to our customers because we can serve them in a different way”. Lastly, we found that resource-seeking advantages such as labour and raw materials were not prominent, constituting only 8% of the total drivers for reshoring.

Table 4. Decision factors served as drivers (D), barriers(B) and enablers(E) in Swedish manufacturing reshoring cases

EP	Decision factors	Quotes	ID1	ID2	ID3	ID4	ID5	ID6	ID7	Respondents							
										ID8	ID9	ID10	ID11	ID12	ID13	ID14	ID15
Resource-seeking	Labour resources	33	D/B	D	D	B/E			D/B/E	B/E	E	D	B				
	Production foundation	5	D/B	D	D		E		D							D/B	
	Customer service	12			D	D	D		D	D		D					
	Industrial agglomeration	2	D														
Market-seeking	Labour cost	15		D	D		D	D	D/B	D/B/E	B				D		
	Logistics performance	31	D/B	D	D	D	D	D	D	D/E		D	D				
	Patriotism	7	D	D	D				D	D		D			D		D
	Raw material access and cost	4				D/B				D							
Efficiency-seeking	Service facilities	2					E			E							
	Controllability of production	32	D	D		D	D	D	D	D	D			D		D	D
	Coordination cost	8	D		D/E				D					D			D
	Delivery lead time	36	D/B	D	D	D	D	D	D/E	D		D	D	D		D	
	External communication	17		D		D/E			D			D	D	D		D	D
	Function synchronisation	28			D	D/E	D/E	D	D	D/B/E	D/E	D	D	D		D	D
	Geographical and cultural distance	22	D	D	D/E	D/E	E	D	D	D							
	Global economic conditions	1							D								
	IT	1	D														
	Inventory level and cost	19	D/B	D/E	D				B		B/E	D	D	D		D	D
	Labour market flexibility	9									D/B	B		B		D	
	Labour productivity	3		D													
	Macroeconomics	3								D			D				
	Manufacturing capacity	30		E		B/E	D	E	B		D/B/E	D/E	D	D		B/E	
	Manufacturing cost	25	D/B			B/E		D	D	D	D	D	D	D	D	D	B
	Manufacturing risks	15			D	D	D	D	D	D	D/B	D/B		D	D	D	
Overhead cost	22		B	D	B/E		D	D/B	D/B	D/B	D/B	D/B	D/B	D			
Political stability	2	D								E							
Production flexibility	29		D		D	D	D	D	D		D	D	D	D	D	D	
Replenishment lead time	6			D			D			D		D	D		D		
Social/ethical concerns	1									D							
Subsidies	1	D															
Supply chain flexibility	4				D		D					D	D				
Supply chain risks	20	D	D	D	D	D	D	D	D	D	D	D	D	D	D	D	D

(continued)

(continued)

Table 4. Continued

EP	Decision factors	Quotes	Respondents															
			ID1	ID2	ID3	ID4	ID5	ID6	ID7	ID8	ID9	ID10	ID11	ID12	ID13	ID14	ID15	ID16
Strategic asset-seeking	Supply chain visibility	3	D															
	Time to market	15	D	D	D	D/B		D	D/E	D			D	D		D	D	D
	Total cost	55	D/B		D	B		D	D	D/B/E	D/E	D		D	D	D/B	D	D
	Trade and payment terms	2					D			D								
	Logistics cost	13				D	D	D	D	D		D	D	D				
	Competitive pressure	6					D	D	D	D	D							
	Competitive priorities	12	E		E	E	D	D	D	D	D	D						
	Core competencies	18				D/B		D/E	D	D/B/E	D/E	D				D		D
	Brand image and reputation	7	D			D				D								D
	Innovation ability	2	D				D											
	Production and process quality	12	D			D	E		D/E		E							D
	Responsible supply chain	4							D		D							
Hybrid-seeking	Servitisation strategy	1																
	Strategy shift	5	E	E			D		E									
	Technology agglomeration	1								D								
	Ownership-related issues	2	B							B								
	Strategic flexibility	2							E		E							
	Currency exchange rate	13	D/B	D			D	D		D/B						D	D	D
	Customisation strategy	5							D		D/B	D/E					D	
	Business partners	18	D/B/E		E	B			D	D/B	D	D/E						
	Customer demand	20	D	D	D/E	D/B/E	E	D	D/E	D		D		D	D	D/B	D	D
	Customer proximity	27		D	D/E	D/E	D/E	D	D/E	D	D/E							
	Facilities and equipment	22	B	E	E	D/B/E		D	B	D/B/E	E	D/B						
	Government incentives	8	D	B/E	E				B	E								
Infrastructure	15	D	D	D		E	D		D/B/E	E	E							
Know-how and IP	33	D/B	D	E	D/E	D/E	D	D	B/E	B	E		D		E	D	D	
Knowledge and technology	13	D		D/E			D		D	D/B		D	D		E	D	D	
Legislation and regulations	18	D	B/E		B	D		B	D/B/E	D	B		D			D	D	
Made-in effect	16	D		D	D				D							D	D	
Management performance	12	D	D		E			D/E	D/B	D/B/E	D						(continued)	

Table 4. Continued

EP	Decision factors	Quotes	ID1	ID2	ID3	ID4	ID5	ID6	ID7	Respondents						ID16
										ID8	ID9	ID10	ID11	ID12	ID13	
	Market opportunities	19		D	D	B/E		D	E	E	E	D	D		D	
	Product quality	44	D	D	D	D	D		D	D	D	D	D		D	D
	R&D	6					D		D	D	D					D
	Responsiveness to market	15	D	D	D	D	D	D		D	D	D	D		D	D
	Supply networks	20	D	E	D	D	D/E	E	D	D/B/E	D/E					
	Sustainable supply chain	15				D/B/E	D	D		D	D	D				D
	Manufacturing automation level	11	B/E		E				B/E	E	B					
Source(s): Authors' own work																

4.2 Reshoring barriers

It's notable that companies' reshoring efforts are primarily driven by a desire to enhance production efficiency. Interestingly, a significant portion (44%, $n = 14$) of reported reshoring barriers also falls within the efficiency-seeking category. Companies encountered challenges due to strict layoff regulations in their offshored locations, such as Poland and China. The inflexibility of these labour markets led to production and investments becoming "locked", leading to substantial exit costs (as highlighted in ID7, ID8, ID9 and ID11). Additionally, high overhead costs (mentioned in seven quotes) and manufacturing costs (mentioned in six quotes) created extra difficulties for production in Sweden. The elevated "total cost" (mentioned in seven quotes) posed a significant obstacle to achieving cost-efficient strategies. Production managers observed that cost estimates made during the reshoring decision-making stage were conservative, resulting in considerably higher actual expenses for factories engaged in reshoring (as indicated by ID1, ID4 and ID15).

Hybrid-seeking barriers accounted for 22% ($n = 7$) of the total barriers, indicating that the same barrier hinders reshoring cases from various perspectives in different reshoring cases. One common example is the availability of "facilities and equipment" in Sweden. Managers stated that they faced challenges in bringing essential tools and machinery back (ID1, ID8 and ID11). Some managers (ID4 and ID7) also mentioned difficulties in acquiring qualified production tools at reasonable prices, while others struggled to maximise the use of existing machines and equipment (ID1 and ID11). Furthermore, Swedish regulations related to employment (ID8 and ID11) made accessing the labour market challenging, and environmental regulations within the Nordic countries necessitated significant adaptations in production processes (ID2, ID7 and ID8), which posed extra challenges for companies. Specific "know-how and IPs" required for production prolonged the strategic asset-seeking and resource-seeking goals. One manager (ID1) noted delays in reshoring initiatives due to crucial components containing IPs from previous suppliers, while another manager (ID8) stated a lack of specific production knowledge in Sweden.

There are also barriers (16% of the total, $n = 5$) that impede companies from effectively pursuing their market-seeking strategies. One significant issue was "logistics performance", as the manager (ID8) explained: "If we move one product [reshoring], our other products might lose priority and that will create delivery issues". Also, the market opportunities in the Nordic region show less potential compared to previous offshored locations. One manager (ID4) noted: "The market is important. It's better that manufacturing is located where the market is. ... the Nordic market is not huge compared to the Asian market". Additionally, increased labour costs (mentioned in seven quotes) resulted in higher product prices, thereby limiting companies' access to additional sales opportunities (ID7, ID8 and ID9).

Some reshoring barriers hinder companies from accessing or leveraging production-related resources and strategic assets in Sweden. Companies found it challenging to hire suitable labour with specific production techniques or skills, such as sewing and tapestry (ID1, ID4 and ID7), indicating labour resources were a disadvantage for Swedish manufacturing. The higher proportion of highly educated personnel in the Nordic countries, compared to developing regions, further complicated the search for qualified personnel. During the reshoring process, companies also experience complex ownership issues related to production components, machinery, factories and knowledge. As one manager (ID8) stated: "Patents, for example. When you decide to move back, and at the last moment, you realize you can't, because the supplier has patents on a small part". Another manager (ID1) added: "There can be some [products] that are tied to a specific tool [...] Even if we own the tool, it might not be possible to move it [...]". Reshoring may also fail due to a lack of

capacity or competencies, as explained by the manager (ID4): “It did not work because we did not have the capacity [...] the competency...”.

4.3 Reshoring enablers

Enablers facilitate companies in accomplishing the reshoring. Most identified enablers (34% of the total, $n = 13$) are efficiency-seeking. The feasibility of a reshoring strategy often depends on the existence of spare capacity within domestic factories and inventory (ID2, ID6, ID8, ID9 and ID11). One manager (ID15) explicitly stated that they would not consider any reshoring plans unless there was available capacity. Another crucial efficiency-seeking enabler is “function synchronisation”, which refers to the advantage of having production proximity to various departments, including purchasing, R&D, assembling, warehousing and logistics, which can significantly enhance production efficiency compared to offshoring (ID4, ID5, ID8 and ID9). In addition, reshoring shorter both geographical and cultural distances between the factory, other focal firms and customers. This closer cooperation among organisations and departments facilitates smoother operations and improves overall production performance (ID3, ID4 and ID5).

Similar to drivers and barriers, 26% ($n = 10$) of enablers facilitated the reshoring from multiple seeking perspectives. For example, the well-structured communication and transportation infrastructure in Sweden is essential for building connections with the target markets (ID8). Furthermore, a reliable and stable electrical and production-related infrastructure serves as a crucial resource-seeking enabler, ensuring uninterrupted production operations (ID5, ID8 and ID11). From an efficiency-seeking perspective, using existing manufacturing infrastructure such as buildings and factories, contribute to cost-efficient production (ID8, ID9 and ID11). Given the challenges of hiring qualified and long-term employees at a relatively lower “labour cost”, maintaining a high level of automation is crucial in the Swedish manufacturing context (ID2, ID4, ID6, ID8, ID9 and ID11). Thus, the “manufacturing automation level” was an important resource for companies to achieve cost-saving objectives (efficiency-seeking enabler) while enhancing production flexibility (strategic asset-seeking enabler).

Strategic asset-seeking enablers comprising 18% of the total ($n = 7$) facilitate companies to align their reshoring strategies with overall development goals. These include maintaining high-quality production and processes in the home country, having the ability to undergo a “strategy shift” and exhibiting “strategic flexibility”. One manager (ID2) stated: “Through lean [strategy], we improved the business, removed inventories, which we could move to the place where we previously had inventories [Sweden]”, and another (ID7) added: “We have adapted the business [strategy] and removed waste [...] which has a positive impact on bringing production back home”.

The rest enablers, each representing 11% ($n = 4$) of the total, provide resource-seeking and market-seeking advantages. While the availability of skilled labour was previously discussed as a reshoring barrier in the Nordic countries, companies emphasised that having access to it is crucial for a smooth and successful reshoring process (highlighted in 11 quotes). One production manager (ID4) emphasised the importance of retaining qualified labour, stating: “...to keep it (reshoring)? Include everyone’s competencies and experience.” Similarly, another manager (ID9) underscored the role of the production team, stating: “What makes this possible? It is the project team you’ve built [...] It’s so important what type of team you’re building”. The availability and accessibility of essential facilities and equipment were also crucial for the smooth implementation of reshoring projects, facilitating the transition and operational efficiency (as mentioned in seven quotes).

5. Discussion

Compared to previous research primarily focused on discovering reshoring drivers or motivations (e.g. [Johansson et al., 2019](#); [Martínez-Mora and Merino, 2020](#)), this paper provides a more comprehensive analysis. By presenting one factor from three different perspectives: drivers, barriers and enablers, it offers a more holistic understanding of reshoring decision-making. After this thorough research, an interesting alignment has emerged, the factors influencing the Nordic reshoring landscape now mirror those commonly reported within the global manufacturing realm. Such findings shed light on factors that influence companies' reshoring decisions, and how these factors can be transferred within various contexts. Based on the four types of strategic-seeking advantages, it also encourages a more in-depth exploration and discussion of the interconnectedness and common experiences within various manufacturing environments.

Turning to the RQs, this research first addressed *RQ1*, which aimed to identify the drivers, barriers and enablers for bringing manufacturing back to Nordic countries. The research revealed key reshoring drivers such as logistics costs and performance, delivery lead time, as well as product and production quality. These findings align with previous research within the Nordic context (e.g. [Engström et al., 2018a](#); [Heikkilä et al., 2018a](#); [Hilletoft et al., 2021](#); [Sequeira et al., 2023](#)). Notably, 23 unique drivers were uncovered, with a strong emphasis on efficiency-seeking motivations, focusing on enhancing the controllability of production, supply chain flexibility and visibility, while reducing inventory level, inventory costs and lowering the total cost. Reshoring cases in Denmark proved that reshoring leads to lower labour productivity ([Stentoft et al., 2018](#)), in contrast, the analysis shows that the "labour productivity" considerations in the Swedish context have instead encouraged managers to reshore (ID2). Additionally, a noteworthy finding was that the prominence of patriotism, and social responsibility were motivating owners to move back, with half of the participants mentioning these aspects. This sense of national pride played a significant role in decision-making. A manager (ID2) stated: "She [the owner] cares about the society here". Another manager (ID8) added: "[the reason for reshoring] Almost patriotic - that we keep the activity [in Sweden]".

The most prevalent barriers to Nordic reshoring include the lack of qualified labour resources with reasonable costs, the availability and affordability of essential facilities and equipment for production. As well as some cost-related factors, such as high labour costs, unpredictable overhead costs and miscalculated total costs. Still, even though companies were driven most by improving productivity-enhance or cost-effective production, most barriers companies reported also impede them in achieving this goal with reshoring. These findings are partially aligned with previous studies ([Arlbjørn and Mikkelsen, 2014](#); [Engström et al., 2018a, 2018b](#); [Stentoft et al., 2016](#)), while this research revealed 27 unique barriers. Our findings suggest that contemporary challenges faced by Nordic companies align more closely with those reported globally. However, this does not imply these barriers did not affect previous Nordic companies' reshoring, it is due to only a few papers having investigated this area in the past. By contrast, our study did not show the importance of "knowledge transfer" in reshoring, which is extensively discussed in both [Nujen et al. \(2019\)](#) and [Fjellström et al. \(2023\)](#).

This paper highlights enablers that increase the success rate of reshoring, identifying key enablers such as manufacturing capacity, labour resources, know-how and IP, infrastructure and automation level. Among these, 30 enablers are newly identified by this research. Notably, the availability of qualified labour was cited as a crucial enabler. Managers emphasised the significance of strong cooperation with staffing agencies and local unions, highlighting the role of collaborative efforts in reshoring success. Manager (ID7) stated: "We

have good cooperation with staffing agencies, and we have a good working climate with the local union to find solutions”. Another manager (ID9) added: “Who is it that makes this possible? It is the project team. The team you’ve built”.

Compared to reshoring cases in the global context, we identified factors such as servitisation, customisation, management performance and technology agglomeration, which align with reshoring findings within the USA, the UK and Polish manufacturing context (Theyel *et al.*, 2018; Moradlou *et al.*, 2021, 2022; Ocicka, 2016). While, the barrier “energy cost” that was highlighted within the US and Spanish contexts (Martínez-Mora and Merino, 2020; Pearce, 2014) did not arise in the findings. Information transfer emerged as a barrier for UK reshoring cases: the break in information flow was considered a significant problem, leading to higher reshoring failure rates (Huq *et al.*, 2016). However, this study did not find quotes related to this. Similarly, factors such as “natural disasters”, “natural resources”, “supply chain resilience” and “tax advantages” did not emerge in Swedish reshoring cases, despite they have been found as important drivers or enablers in the US and the UK context (Tate, 2014; Ellram *et al.*, 2013; Srari and Ané, 2016; Moradlou *et al.*, 2021).

The focus now shifts towards RQ2, which focuses on discovering the linkage among these drivers, barriers and enablers based on the eclectic paradigm. Several findings have supported the discussion that drivers, barriers and enablers are transferable based on specific cases (Engström *et al.*, 2018a, 2018b). Among these, factors related to the development of digitalisation and sustainability are presented in multiple perspectives with new roles. Previous research emphasised that increased automation level in the Nordic country is a major reshoring motivation (El-Sahli and Gullstrand, 2023), especially in the Danish manufacturing context (Arlbjørn and Mikkelsen, 2014). While this did not emerge as a driver in the Swedish cases, it was cited as a reshoring barrier and enabler. The decisive impact of automation in addressing the challenges posed by high labour costs. Manager (ID1) stated: “The development of automatisisation at the companies is very decisive to where manufacturing can be placed. Our production is no longer that manpower intensive” and “The day we can automate assembly, we’ll move everything home”. When mentioning the timing for moving back, the manager (ID3) added: “If there were loans or very beneficial support from the government to invest in automatisisation equipment to take manufacturing home”. The potential for government support in automation investments was also emphasised, indicating a clear link between policy initiatives and reshoring decisions.

Sustainable development is another major trend in the current and future European manufacturing industry (McIvor *et al.*, 2025; Martínez-Mora and Merino, 2020). With the target of zero emissions by 2050, companies are motivated to build more sustainable operations. The current study found that sustainability is increasingly becoming a key driver, this study has revealed it through the specific lens of sustainable supply chains. One manager (ID5) has witnessed sustainability concerns gaining prominence: “The part about the environment is becoming stronger, mainly the big firms have pushed themselves on this”. Sustainability functions as both a strategic asset- and resource-seeking driver, managers reshoring to reduce carbon footprints by building a more local supply chain and choosing suppliers who have the same sustainable business strategy, which is also very attractive to specific customers (ID4, ID6, ID8, ID9, ID11, ID16). However, the study also found that stringent environmental regulations in Nordic countries could act as a barrier, as one manager (ID4) pointed out: “It is not environmentally suitable to produce in Sweden and send to China [...] some production will be located in Asia or other continents”. This highlights the necessity for companies to ensure that their sustainable supply chains comply with both production regulations and customer expectations before initiating the reshoring process. Such compliance can significantly increase the likelihood of successful reshoring.

Compared to RBV and TCE, the application of the eclectic paradigm enables scholars to capture the complexities of reshoring decisions and analyses factors with a more holistic approach. A recent reshoring study by [McIvor et al. \(2025\)](#) applied a framework that integrated the natural RBV and TCE to analyse the German automotive industry, for a better integration for investigating sustainability and economic perspectives. The study was interesting, but factors were only discussed from the drivers' perspective, which missed the opportunities to provide a holistic view. We established the framework that encourages the application of the eclectic paradigm in analysis reshoring related phenomena to scholars in future research. By not only highlighting four types of strategic-seeking categories, but integrating the three perspectives of one decision factor (i.e. driver, barrier and enabler perspective).

6. Conclusions

6.1 Theoretical contribution

This research is one of the few studies integrating strategic advantages of the eclectic paradigm with the three aspects of decision factors in the analysis of manufacturing reshoring. This study offers fresh insights into reshoring within the Nordic manufacturing sector, uncovering key contextual dynamics that influence decision-making. By focusing on the broader patterns and relationships, this research provides a more nuanced and engaging exploration of the reshoring of decision-making, particularly with the eclectic paradigm. Previously overlooked aspects, such as patriotism, customisation strategies and production control, were revealed in this research, thus deepening our understanding of what drives companies to bring operations back home and what unique barriers and enablers shape these efforts.

What sets this study apart is its holistic approach, rather than examining these factors separately, it explores the interconnectedness of drivers, barriers and enablers. Drawing on the eclectic paradigm, the findings highlight the dominance of efficiency-seeking drivers, aligning with earlier studies (e.g. [Hilletoft et al., 2021](#); [Sequeira et al., 2023](#)). This integrated perspective enriches the theoretical framework and offers practical insights for companies looking to optimise their reshoring strategies. By applying the eclectic paradigm to the reshoring phenomenon, it again successfully proves the applicability of the paradigm beyond traditional foreign direct investment contexts ([Li et al., 2025](#)). The findings integrate the three perspectives of one factor with the four types of strategic-seeking categories. By doing so, it bridges gaps between separate streams of research and provides a clearer understanding of how reshoring decisions unfold in practice. From an academic standpoint, this move encourages the application of the eclectic paradigm, as well as refining it rather than analysis from solely driver perspective.

6.2 Managerial implications

This study offers practical insights into reshoring decision-making content for managers and industry leaders. This understanding empowers them to make well-informed decisions in the future. Managers are able to use the identified drivers in their strategic decision-making, and in that way counteract myopic decision-making. In cases where drivers and barriers are conflicting, it is important for managers to consider how they can offset these conflicts. For example, if manufacturing cost is a barrier due to high salaries and labour-intensive production, managers should see, if factors such as labour productivity and manufacturing automation level can act as enablers. Similarly, factors such as R&D and know-how and IP can help to enable a strategy shift. There is also an increasing awareness of the potential risks inherent in supply chain disruptions ([Moradlou et al., 2021](#)). Adapting to such risks might justify accepting short-term barriers in favour of long-term stability.

6.3 Policy implications

Given the growing interest in strengthening domestic manufacturing among Nordic governments, this study offers valuable insights for policymakers seeking to support reshoring initiatives and enhance national competitiveness. For example, the pivotal role of efficiency-seeking factors in driving manufacturing back, suggesting that governments can focus on these areas to attract more companies. Identifying barriers such as skilled labour shortages and high manufacturing costs emphasises the need for strategic planning from a governmental perspective. Policymakers can collaborate with industries to develop targeted skill development programmes, addressing specific gaps in the local labour market. Furthermore, armed with knowledge about enablers like manufacturing automation and local labour resources, companies and governments can collaborate effectively to enhance reshoring success.

6.4 Limitations and further research

This study uses Swedish manufacturing data while aiming to represent the Nordic manufacturing landscape. Swedish manufacturing has a strong emphasis on innovation, sustainability and global competitiveness. However, an exclusive focus on Sweden might restrict a full depiction of the broader Nordic manufacturing spectrum. The findings may not fully capture variations in other Nordic countries. Several potential research avenues for future research. For example, future research should include diverse Nordic samples for a more comprehensive understanding. Additionally, the eclectic paradigm did not fully cover contingency factors, such as product characteristics, firm-specific attributes and personal experiences, which significantly influence companies' decision-making. Future studies could also enhance the depth of analysis by exploring alternative theoretical frameworks or conducting multi-theory analyses (Yu and Kim, 2018; Boffelli and Johansson, 2020; McIvor and Bals, 2021), providing a more holistic understanding of reshoring determinants. Additionally, scholars can discover the reshoring decision-making content from other countries or regions, contributing to a global understanding of reshoring dynamics. As one key question arises regarding whether reshoring drivers, barriers, and enablers is becoming globally consistent. If similar patterns emerge across different regions, this may indicate an alignment in reshoring determinants, reducing the need for region-specific studies. This can first stress the concern of the general consistent understanding of reshoring decision factors. Furthermore, even if basic factors remain the same globally, it will be interesting to discover the difference between the significance of each factor in different countries (e.g. which factors have major influences in specific locations and how they differ from country to country). Last, but not least, companies of different sizes and industries may experience varying impacts on reshoring decisions, necessitating evaluations of the drivers, enablers and barriers for reshoring.

References

- Ancarani, A., Di Mauro, C., Fratocchi, L., Orzes, G. and Sartor, M. (2015), "Prior to reshoring: a duration analysis of foreign manufacturing ventures", *International Journal of Production Economics*, Vol. 169, pp. 141-155, doi: [10.1016/j.ijpe.2015.07.031](https://doi.org/10.1016/j.ijpe.2015.07.031).
- Arlbjørn, J.S. and Mikkelsen, O.S. (2014), "Backshoring manufacturing: notes on an important but under-researched theme", *Journal of Purchasing and Supply Management*, Vol. 20 No. 1, pp. 60-62, doi: [10.1016/j.pursup.2014.02.003](https://doi.org/10.1016/j.pursup.2014.02.003).
- Ashby, A., Leat, M. and Hudson-Smith, M. (2012), "Making connections: a review of supply chain management and sustainability literature", *Supply Chain Management: An International Journal*, Vol. 17 No. 5, pp. 497-516.

- Baraldi, E., Ciabuschi, F., Fraticchi, L., Pedroletti, D. and Picciotti, A. (2024), "The roles of key suppliers in network formation when reshoring", *Journal of Business and Industrial Marketing*, doi: [10.1108/JBIM-02-2023-0101](https://doi.org/10.1108/JBIM-02-2023-0101).
- Barbieri, P., Ciabuschi, F., Fraticchi, L. and Vignoli, M. (2018), "What do we know about manufacturing reshoring?", *Journal of Global Operations and Strategic Sourcing*, Vol. 11 No. 1, pp. 79-122, doi: [10.1108/JGOSS-02-2017-0004](https://doi.org/10.1108/JGOSS-02-2017-0004).
- Barney, J.B. (1991), "Firm resources and sustained competitive advantage", *Journal of Management*, Vol. 17 No. 1, pp. 49-61.
- Boffelli, A. and Johansson, M. (2020), "What do we want to know about reshoring? Towards a comprehensive framework based on a meta-synthesis", *Operations Management Research*, Vol. 13 Nos 1/2, pp. 53-69, doi: [10.1007/s12063-020-00155-y](https://doi.org/10.1007/s12063-020-00155-y).
- Boffelli, A., Orzes, G. and Dotti, S. (2018), "How to reshore": some evidence from the apparel industry", *IEEE Engineering Management Review*, Vol. 46 No. 4, pp. 122-127, doi: [10.1109/EMR.2018.2886183](https://doi.org/10.1109/EMR.2018.2886183).
- Boffelli, A., Fraticchi, L., Kalchschmidt, M. and Silva, S. (2021), "Doing the right thing or doing things right: what is better for a successful manufacturing reshoring?", *Operations Management Research*, Vol. 14 Nos 1/2, pp. 1-16, doi: [10.1007/s12063-021-00183-2](https://doi.org/10.1007/s12063-021-00183-2).
- Boffelli, A., Golini, R., Orzes, G. and Dotti, S. (2020), "Open the box: a behavioural perspective on the reshoring decision-making and implementation process", *Journal of Purchasing and Supply Management*, Vol. 26 No. 3, p. 100623, doi: [10.1016/j.pursup.2020.100623](https://doi.org/10.1016/j.pursup.2020.100623).
- Chen, H., Hsu, C., Shih, Y. and Caskey, D. (2022), "The reshoring decision under uncertainty in the post-COVID-19 era", *Journal of Business and Industrial Marketing*, Vol. 37 No. 10, pp. 2064-2074, doi: [10.1108/JBIM-01-2021-0066](https://doi.org/10.1108/JBIM-01-2021-0066).
- Cook, D.J., Greengold, N.L., Ellrodt, A.G. and Weingarten, S.R. (1997), "The relation between systematic reviews and practice guidelines", *Annals of Internal Medicine*, Vol. 127 No. 3, pp. 403-418.
- D'Ambrosio, A. and Lavoratori, K. (2025), "Reshoring to survive? The other side of de-globalization", *J. Ind. Bus. Econ.*, pp. 1-23, doi: [10.1007/s40812-025-00342-7](https://doi.org/10.1007/s40812-025-00342-7).
- Dachs, B., Kinkel, S., Jäger, A. and Palčič, I. (2019), "Backshoring of production activities in European manufacturing", *Journal of Purchasing and Supply Management*, Vol. 25 No. 3, p. 100531, doi: [10.1016/j.pursup.2019.02.003](https://doi.org/10.1016/j.pursup.2019.02.003).
- Dunning, J.H. (1983), *International Production and the Multinational Enterprise (RLE International Business)*, Routledge, London, UK.
- Dunning, J.H. (1988), "The eclectic paradigm of international production: a restatement and some possible extensions", *Journal of International Business Studies*, Vol. 19 No. 1, pp. 1-31, available at: www.jstor.org/stable/154984
- Dunning, J.H. (1998), "Location and the multinational enterprise: a neglected factor?", *Journal of International Business Studies*, Vol. 29 No. 1, pp. 45-66, doi: [10.1057/palgrave.jibs.8490024](https://doi.org/10.1057/palgrave.jibs.8490024).
- Eisenhardt, K.M. (1989), "Building theories from case study research", *The Academy of Management Review*, Vol. 14 No. 4, pp. 532-550, doi: [10.2307/258557](https://doi.org/10.2307/258557).
- Ellram, L.M. (1996), "The use of case study method in logistics research", *Journal of Business Logistics*, Vol. 17 No. 2, pp. 93-138.
- Ellram, L.M., Tate, W.L. and Petersen, K.J. (2013), "Offshoring and reshoring: an update on the manufacturing location decision", *Journal of Supply Chain Management*, Vol. 49 No. 2, pp. 14-22, doi: [10.1111/jscm.12019](https://doi.org/10.1111/jscm.12019).
- El-Sahli, Z. and Gullstrand, J. (2023), "Why reshore? Evidence from Swedish firms", *Economics Letters*, Vol. 227, p. 111116, doi: [10.1016/j.econlet.2023.111116](https://doi.org/10.1016/j.econlet.2023.111116).
- Engström, G., Hilletofth, P., Eriksson, D. and Sollander, K. (2018b), "Drivers and barriers of reshoring in the Swedish manufacturing industry", *World Review of Intermodal Transportation Research*, Vol. 7 No. 3, pp. 195-220, doi: [10.1504/WRITR.2018.093550](https://doi.org/10.1504/WRITR.2018.093550).

- Engström, G., Sollander, K., Hilletoft, P. and Eriksson, D. (2018a), "Reshoring drivers and barriers in the Swedish manufacturing industry", *Journal of Global Operations and Strategic Sourcing*, Vol. 11 No. 2, pp. 174-201, doi: [10.1108/JGOSS-06-2017-0014](https://doi.org/10.1108/JGOSS-06-2017-0014).
- Eriksson, D. and Engström, A. (2021), "Using critical realism and abduction to navigate theory and data in operations and supply chain management research", *Supply Chain Management: An International Journal*, Vol. 26 No. 2, pp. 224-239, doi: [10.1108/SCM-03-2020-0091](https://doi.org/10.1108/SCM-03-2020-0091).
- Eriksson, D., Hilletoft, P., Tate, W. and Gothager, M. (2021), "Critical manufacturing prerequisites for successful reshoring", *Operations and Supply Chain Management: An International Journal*, Vol. 14 No. 2, pp. 249-260.
- Fjellström, D., Hilletoft, P., Fang, T., Kumar, V., Kumar, A. and Tan, K. (2023), "Manufacturing relocation ambiguity model: a prerequisite for knowledge management", *British Journal of Management*, Vol. 34 No. 3, pp. 1100-1116, doi: [10.1111/1467-8551.12710](https://doi.org/10.1111/1467-8551.12710).
- Fratocchi, L., Ancarani, A., Barbieri, P., Di Mauro, C., Nassimbeni, G., Sartor, M. and Zaroni, A. (2016), "Motivations of manufacturing reshoring: an interpretative framework", *International Journal of Physical Distribution and Logistics Management*, Vol. 46 No. 2, pp. 98-127, doi: [10.1108/IJPDLM-06-2014-0131](https://doi.org/10.1108/IJPDLM-06-2014-0131).
- Grappi, S., Romani, S. and Bagozzi, R.P. (2020), "The effects of reshoring decisions on employees", *Personnel Review*, Vol. 49 No. 6, pp. 1254-1268, doi: [10.1108/PR-12-2018-0482](https://doi.org/10.1108/PR-12-2018-0482).
- Gylling, M., Heikkilä, J., Jussila, K. and Saarinen, M. (2015), "Making decisions on offshore outsourcing and backshoring: a case study in the bicycle industry", *International Journal of Production Economics*, Vol. 162, pp. 92-100, doi: [10.1016/j.ijpe.2015.01.006](https://doi.org/10.1016/j.ijpe.2015.01.006).
- Heikkilä, J., Martinsuo, M. and Nenonen, S. (2018a), "Backshoring of production in the context of a small and open Nordic economy", *Journal of Manufacturing Technology Management*, Vol. 29 No. 4, pp. 658-675, doi: [10.1108/JMTM-12-2016-0178](https://doi.org/10.1108/JMTM-12-2016-0178).
- Heikkilä, J., Nenonen, S., Olhager, J. and Stentoft, J. (2018b), "Manufacturing relocation abroad and back: empirical evidence from the Nordic countries", *World Review of Intermodal Transportation Research*, Vol. 7 No. 3, pp. 221-240, doi: [10.1504/WRITR.2018.093563](https://doi.org/10.1504/WRITR.2018.093563).
- Hilletoft, P., Sequeira, M. and Adlemo, A. (2019), "Three novel fuzzy logic concepts applied to reshoring decision-making", *Expert Systems with Applications*, Vol. 126, pp. 133-143, doi: [10.1016/j.eswa.2019.02.018](https://doi.org/10.1016/j.eswa.2019.02.018).
- Hilletoft, P., Sequeira, M. and Tate, W. (2021), "Fuzzy-logic-based support tools for initial screening of manufacturing reshoring decisions", *Industrial Management and Data Systems*, Vol. 121 No. 5, pp. 965-992, doi: [10.1108/IMDS-05-2020-0290](https://doi.org/10.1108/IMDS-05-2020-0290).
- Hilmola, O.P., Li, H., Hilletoft, P. and Fjellström, D. (2025), "Evaluation of EU-27 countries as manufacturing location: strengths and weaknesses model", *Operations Management Research*, doi: [10.1007/s12063-024-00539-4](https://doi.org/10.1007/s12063-024-00539-4).
- Huq, F., Pawar, K.S. and Rogers, H. (2016), "Supply chain configuration conundrum: how does the pharmaceutical industry mitigate disturbance factors?", *Production Planning and Control*, Vol. 27 No. 14, pp. 1206-1220, doi: [10.1080/09537287.2016.1193911](https://doi.org/10.1080/09537287.2016.1193911).
- Johansson, M. and Olhager, J. (2018a), "Manufacturing relocation through offshoring and backshoring: the case of Sweden", *Journal of Manufacturing Technology Management*, Vol. 29 No. 4, pp. 637-657, doi: [10.1108/JMTM-01-2017-0006](https://doi.org/10.1108/JMTM-01-2017-0006).
- Johansson, M. and Olhager, J. (2018b), "Comparing offshoring and backshoring: the role of manufacturing site location factors and their impact on post-relocation performance", *International Journal of Production Economics*, Vol. 205, pp. 37-46, doi: [10.1016/j.ijpe.2018.08.027](https://doi.org/10.1016/j.ijpe.2018.08.027).
- Johansson, M., Olhager, J., Heikkilä, J. and Stentoft, J. (2019), "Offshoring versus backshoring: empirically derived bundles of relocation drivers, and their relationship with benefits", *Journal of Purchasing and Supply Management*, Vol. 25 No. 3, p. 100509, doi: [10.1016/j.pursup.2018.07.003](https://doi.org/10.1016/j.pursup.2018.07.003).

- Joubioux, C. and Vanpoucke, E. (2016), "Towards right-shoring: a framework for off-and re-shoring decision making", *Operations Management Research*, Vol. 9 Nos 3/4, pp. 117-132, doi: [10.1007/s12063-016-0115-y](https://doi.org/10.1007/s12063-016-0115-y).
- Lamperti, F., Lavoratori, K. and Tredicine, L. (2025), "From globalization to reshoring? The role of industry 4.0 in global supply chains across Europe", *Economics of Innovation and New Technology*, pp. 1-23, doi: [10.1080/10438599.2025.2487145](https://doi.org/10.1080/10438599.2025.2487145).
- Leisner, P. and Nielsen, L. (2019), "Offshoring and backshoring of surface finishing from the perspective of the Nordic countries", *Transactions of the IMF*, Vol. 97 No. 2, pp. 54-56, doi: [10.1080/00202967.2019.1571261](https://doi.org/10.1080/00202967.2019.1571261).
- Li, H., Hilletofth, P., Eriksson, D. and Tate, W. (2025), "Understanding the manufacturing reshoring decision-making content through the lens of the eclectic paradigm: a systematic literature review", *European Business Review*, Vol. 37 No. 1, pp. 16-48, doi: [10.1108/EBR-10-2023-0300](https://doi.org/10.1108/EBR-10-2023-0300).
- Liu, F., Liu, C., Wang, X., Park, K. and Fang, M. (2023), "Keep concentrated and carry on: redesigning supply chain concentration in the face of COVID-19", *International Journal of Logistics Research and Applications*, Vol. 27 No. 10, pp. 1681-1704, doi: [10.1080/13675567.2023.2175803](https://doi.org/10.1080/13675567.2023.2175803).
- Lund, H. and Steen, M. (2020), "Make at home or abroad? Manufacturing reshoring through a GPN lens: a Norwegian case study", *Geoforum*, Vol. 113, pp. 154-164, doi: [10.1016/j.geoforum.2020.04.015](https://doi.org/10.1016/j.geoforum.2020.04.015).
- McIvor, R. and Bals, L. (2021), "A multi-theory framework for understanding the reshoring decision", *International Business Review*, Vol. 30 No. 6, p. 101827, doi: [10.1016/j.ibusrev.2021.101827](https://doi.org/10.1016/j.ibusrev.2021.101827).
- McIvor, R., Bals, L., Dereymaeker, T. and Foerstl, K. (2025), "Integrating sustainability and economic perspectives in reshoring: insights from the German automotive industry", *Supply Chain Management: An International Journal*, Vol. 30 No. 2, doi: [10.1108/SCM-12-2023-0640](https://doi.org/10.1108/SCM-12-2023-0640).
- Magnani, G. and Gioia, D. (2023), "Using the Gioia methodology in international business and entrepreneurship research", *International Business Review*, Vol. 32 No. 2, p. 102097, doi: [10.1016/j.ibusrev.2022.102097](https://doi.org/10.1016/j.ibusrev.2022.102097).
- Malhotra, M.K. and Grover, V. (1998), "An assessment of survey research in POM: from constructs to theory", *Journal of Operations Management*, Vol. 16 No. 4, pp. 407-425, doi: [10.1016/S0272-6963\(98\)00021-7](https://doi.org/10.1016/S0272-6963(98)00021-7).
- Martínez-Mora, C. and Merino, F. (2020), "Consequences of sustainable innovations on the reshoring drivers' framework", *Journal of Manufacturing Technology Management*, Vol. 31 No. 7, pp. 1373-1390, doi: [10.1108/JMTM-12-2019-0426](https://doi.org/10.1108/JMTM-12-2019-0426).
- Mohiuddin, M., Rashid, M., Al Azad, M. and Su, Z. (2019), "Back-shoring or re-shoring: determinants of manufacturing offshoring from emerging to least developing countries (LDCs)", *International Journal of Logistics Research and Applications*, Vol. 22 No. 1, pp. 78-97, doi: [10.1080/13675567.2018.1475554](https://doi.org/10.1080/13675567.2018.1475554).
- Moradlou, H., Fratocchi, L., Skipworth, H. and Ghadge, A. (2022), "Post-Brexit back-shoring strategies: what UK manufacturing companies could learn from the past?", *Production Planning and Control*, Vol. 33 No. 14, pp. 1319-1336, doi: [10.1080/09537287.2020.1863500](https://doi.org/10.1080/09537287.2020.1863500).
- Moradlou, H., Reefke, H., Skipworth, H. and Roscoe, S. (2021), "Geopolitical disruptions and the manufacturing location decision in multinational company supply chains: a Delphi study on Brexit", *International Journal of Operations and Production Management*, Vol. 41 No. 2, pp. 102-130, doi: [10.1108/IJOPM-07-2020-0465](https://doi.org/10.1108/IJOPM-07-2020-0465).
- Moretto, A., Patrucco, A. and Harland, C. (2020), "The dynamics of reshoring decisions and the role of purchasing", *International Journal of Production Research*, Vol. 58 No. 19, pp. 5929-5944, doi: [10.1080/00207543.2019.1661534](https://doi.org/10.1080/00207543.2019.1661534).
- Nujen, B.B., Mwesummo, D., Solli-Sæther, H., Slyngstad, A. and Halse, L.L. (2019), "Backshoring readiness", *Journal of Global Operations and Strategic Sourcing*, Vol. 12 No. 1, pp. 172-195, doi: [10.1108/JGOSS-05-2018-0020](https://doi.org/10.1108/JGOSS-05-2018-0020).
- Ocicka, B. (2016), "Reshoring: implementation issues and research opportunities", *Management*, Vol. 20 No. 2, pp. 103-117, doi: [10.1515/manment-2015-0053](https://doi.org/10.1515/manment-2015-0053).

- Olhager, J. and Harfeldt-Berg, M. (2025), "Reshoring before and during the COVID-19 pandemic in an advanced economy", *International Journal of Physical Distribution and Logistics Management*, Vol. 55 No. 11, pp. 1-22, doi: [10.1108/IJPDLM-11-2023-0440](https://doi.org/10.1108/IJPDLM-11-2023-0440).
- Pearce, J.A. (2014), "Why domestic outsourcing is leading America's reemergence in global manufacturing", *Business Horizons*, Vol. 57 No. 1, pp. 27-36, doi: [10.31387/oscm0280186](https://doi.org/10.31387/oscm0280186).
- Pedroletti, D. and Ciabuschi, F. (2023), "Reshoring: a review and research agenda", *Journal of Business Research*, Vol. 164, p. 114005, doi: [10.1016/j.jbusres.2023.114005](https://doi.org/10.1016/j.jbusres.2023.114005).
- Pegoraro, D., De Propriis, L. and Chidlow, A. (2022), "Regional factors enabling manufacturing reshoring strategies: a case study perspective", *Journal of International Business Policy*, Vol. 5 No. 1, pp. 112-133, doi: [10.1057/s42214-021-00112-x](https://doi.org/10.1057/s42214-021-00112-x).
- Pla-Barber, J., Pallás-Rocafull, A., Botella-Andreu, A. and Villar, C. (2025), "Reshoring and regionalization of global value chains: evidence in manufacturing sectors", *Global Value Chains and Competitiveness*, Palgrave Macmillan, Cham, doi: [10.1007/978-3-031-78308-1_6](https://doi.org/10.1007/978-3-031-78308-1_6).
- Presley, A., Meade, L. and Sarkis, J. (2016), "A strategic sourcing evaluation methodology for reshoring decisions", *Supply Chain Forum: An International Journal*, Vol. 17 No. 3, pp. 156-169, doi: [10.1080/16258312.2016.1215852](https://doi.org/10.1080/16258312.2016.1215852).
- Sequeira, M., Adlemo, A. and Hilletoft, P. (2023), "A hybrid fuzzy-AHP-TOPSIS model for evaluation of manufacturing relocation decisions", *Operations Management Research*, Vol. 16 No. 1, pp. 164-191, doi: [10.1007/s12063-022-00284-6](https://doi.org/10.1007/s12063-022-00284-6).
- Sequeira, M., Hilletoft, P. and Adlemo, A. (2021), "AHP-based support tools for initial screening of manufacturing reshoring decisions", *Journal of Global Operations and Strategic Sourcing*, Vol. 14 No. 3, pp. 502-527, doi: [10.1108/JGOSS-07-2020-0037](https://doi.org/10.1108/JGOSS-07-2020-0037).
- Srai, J.S. and Ané, C. (2016), "Institutional and strategic operations perspectives on manufacturing reshoring", *International Journal of Production Research*, Vol. 54 No. 23, pp. 7193-7211, doi: [10.1080/00207543.2016.1193247](https://doi.org/10.1080/00207543.2016.1193247).
- Stentoft, J., Mikkelsen, O.S. and Jensen, J.K. (2016), "Offshoring and backshoring manufacturing from a supply chain innovation perspective", *Supply Chain Forum: An International Journal*, Vol. 17 No. 4, pp. 190-204, doi: [10.1080/16258312.2016.1239465](https://doi.org/10.1080/16258312.2016.1239465).
- Stentoft, J., Mikkelsen, O.S., Jensen, J.K. and Rajkumar, C. (2018), "Performance outcomes of offshoring, backshoring and staying at home manufacturing", *International Journal of Production Economics*, Vol. 199, pp. 199-208, doi: [10.1016/j.ijpe.2018.03.009](https://doi.org/10.1016/j.ijpe.2018.03.009).
- Tate, W.L. (2014), "Offshoring and reshoring: US insights and research challenges", *Journal of Purchasing and Supply Management*, Vol. 20 No. 1, pp. 66-68, doi: [10.1016/j.pursup.2014.01.007](https://doi.org/10.1016/j.pursup.2014.01.007).
- Theyel, G. and Hofmann, K. (2021), "Manufacturing location decisions and organizational agility", *Multinational Business Review*, Vol. 29 No. 2, pp. 166-188, doi: [10.1108/MBR-02-2020-0035](https://doi.org/10.1108/MBR-02-2020-0035).
- Theyel, G., Hofmann, K. and Gregory, M. (2018), "Understanding manufacturing location decision making: rationales for retaining, offshoring, reshoring, and hybrid approaches", *Economic Development Quarterly*, Vol. 32 No. 4, pp. 300-312, doi: [10.1177/0891242418800222](https://doi.org/10.1177/0891242418800222).
- Thomas, C.G. (2021), *Research Methodology and Scientific Writing*, 2nd ed, Springer, Cham, doi: [10.1007/978-3-030-64865-7](https://doi.org/10.1007/978-3-030-64865-7).
- Treven, P. (2022), "Reshoring manufacturing: how digital transformation is bringing back jobs", *Ericsson*, available at: www.ericsson.com/en/blog/2022/3/reshoring-manufacturing-how-digital-transformation-is-bringing-back-jobs (accessed 9 March 2025).
- Uluskan, M., Godfrey, A.B. and Joines, J.A. (2017), "Impact of competitive strategy and cost-focus on global supplier switching (reshore and relocation) decisions", *The Journal of the Textile Institute*, Vol. 108 No. 8, pp. 1308-1318, doi: [10.1080/00405000.2016.1245596](https://doi.org/10.1080/00405000.2016.1245596).
- Üner, M.M., Cigdemoglu, C., Wang, Y., Yalcin, A. and Cavusgil, S.T. (2023), "A review of the evolving conceptualization of internationalization from a global value chain perspective", *Review of International Business and Strategy*, Vol. 33 No. 4, pp. 549-580, doi: [10.1108/RIBS-05-2022-0053](https://doi.org/10.1108/RIBS-05-2022-0053).

- Vanchan, V., Mulhall, R. and Bryson, J. (2018), "Repatriation or reshoring of manufacturing to the US and UK: dynamics and global production networks or from here to there and back again", *Growth and Change*, Vol. 49 No. 1, pp. 97-121, doi: [10.1111/grow.12224](https://doi.org/10.1111/grow.12224).
- Vissak, T. (2024), "A literature review on CEE firms' outward internationalization failures: definitions, processes, causes and consequences", *Review of International Business and Strategy*, Vol. 34 No. 5, pp. 632-658, doi: [10.1108/RIBS-12-2023-0145](https://doi.org/10.1108/RIBS-12-2023-0145).
- Voss, C., Tsikriktsis, N. and Frohlich, M. (2002), "Case research in operations management", *International Journal of Operations and Production Management*, Vol. 22 No. 2, pp. 195-219, doi: [10.1108/01443570210414329](https://doi.org/10.1108/01443570210414329).
- Wiesmann, B., Snoei, J.R., Hilletofth, P. and Eriksson, D. (2017), "Drivers and barriers to reshoring: a literature review on offshoring in reverse", *European Business Review*, Vol. 29 No. 1, pp. 15-42, doi: [10.1108/EBR-03-2016-0050](https://doi.org/10.1108/EBR-03-2016-0050).
- Williamson, O.E. (1985), *The Economic Institutions of Capitalism: Firms, Markets and Relational Contracting*, Free Press, New York, NY.
- World Bank (2023), "World development indicator", available at: <https://datacatalog.worldbank.org/search/dataset/0037712/World-Development-Indicators>
- Wu, J. (2025), "The trend of deglobalization: Reshoring, Friend-Shoring, and regionalization", *Global Trends in Manufacturing Supply Chains. Springer Series in Supply Chain Management*, Vol. 26 Springer, Singapore, doi: [10.1007/978-981-96-3228-2_4](https://doi.org/10.1007/978-981-96-3228-2_4).
- Yin, R.K. (2014), *Case Study Research: Design and Methods*, 5th ed, Sage Publications, London.
- Yu, U.J. and Kim, J.H. (2018), "Financial productivity issues of offshore and made-in-USA" through reshoring", *Journal of Fashion Marketing and Management: An International Journal*, Vol. 22 No. 3, pp. 317-334, doi: [10.1108/JFMM-12-2017-0136](https://doi.org/10.1108/JFMM-12-2017-0136).

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